BMEN 600
Midterm Research Plan

Transcriptomic transferability between human brain regions

A machine-learning study of predictive sharing and disease-relevant molecular characteristics

Prepared: October 9, 2026
Midterm deadline: October 16, 2026
Final project deadline: December 4, 2026

Team members: [INSERT ALL 3-4 TEAM MEMBER NAMES]
Working GitHub repository: [INSERT WORKING REPOSITORY URL]

Draft status. The scientific sections include verified preliminary analyses of the supplied GTEx files. Before submission, replace the identity and repository fields, confirm the proposed role assignments, and complete the actual author-contribution and team-verification statements. A functioning repository is a required component of the assignment.

Six pages of main text, excluding this title page and references.

1. Introduction and Motivation

Parkinson's disease, Alzheimer's disease and amyotrophic lateral sclerosis (ALS) affect particular neuronal populations and brain regions unevenly. This selective vulnerability raises a biomedical question: why do some neural systems degenerate while others remain comparatively preserved? Molecular and cellular properties help frame this problem, but regional vulnerability cannot be reduced to one expression pattern [1], [2]. Better characterization of regional molecular organization could support research into disease mechanisms and the design of more informative experimental models.

Obtaining matched transcriptomic measurements from multiple regions of a human brain is difficult. Consequently, it is useful to determine how much information measured in one region can predict expression in another. Brain atlases demonstrate anatomical organization of gene expression [3], [4], while GTEx supplies molecular measurements across tissues from the same donors [5]. A predictive analysis can therefore test whether source-region expression captures variation in target-region expression among people whose data were not used to train the model.

Research question: To what extent can machine learning quantify transcriptomic transferability between human brain regions, and how are differences in transferability associated with molecular characteristics relevant to regional vulnerability in neurological disease?

Here, transferability is operationally defined as held-out-donor prediction performance under a specified model. The project will compare directional region pairs, summarize gene-wise R^2, and investigate associations with regional expression differences, expression variability and a small prespecified set of disease-associated genes. It will contribute a reproducible computational framework for evaluating predictive relationships and generating biological hypotheses, rather than claiming to identify causes of neurodegeneration.

The biomedical engineering relevance lies in assessing when tissue measurements can serve as informative computational proxies and when regional differences limit such inference. Potential end-users include researchers developing transcriptomic models or prioritizing brain regions for follow-up experiments. Patients and caregivers motivate the disease context, but the analysis is not a clinical diagnostic tool or an immediately validated biomarker. Clinical usefulness would require disease cohorts and additional experimental validation.

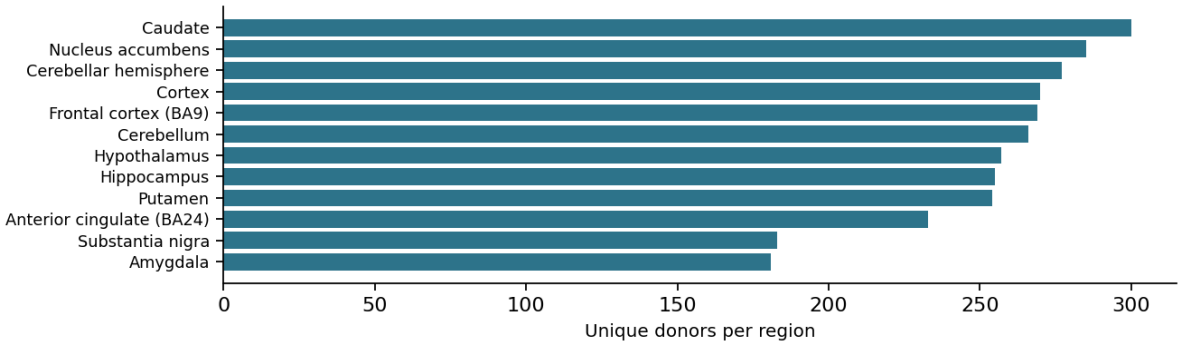


Figure 1. Available unique donors in the 12 brain-region files, calculated from the supplied data. Unequal coverage motivates pair-specific donor matching and cohort-size reporting. The separately uploaded spinal-cord file is excluded from the main analysis.

2. Background and Research Gap

Selective vulnerability provides the biological motivation, but it also defines the limits of expression-based inference. Fu, Hardy and Duff synthesize evidence that neuronal susceptibility reflects multiple intrinsic properties and disease-related processes [1]. In a more focused study, Fu *et al.* combined single-nucleus transcriptomic analyses with experimental investigations of tau accumulation, identifying a tau-homeostasis signature and implicating BAG3-related proteostasis [2]. Together, these studies motivate molecular comparisons while showing that cell-specific and experimental evidence is needed to connect expression with degeneration. Bulk regional measurements alone cannot establish the same mechanisms.

Spatial transcriptomic resources demonstrate that the human brain has structured molecular organization. Hawrylycz *et al.* used anatomically detailed microarray sampling to build an adult human brain atlas [3]. Their work enables comparisons across finely defined anatomical locations, although deep sampling in a small number of donors limits population-level inference. Subsequent analysis identified reproducible regional expression signatures associated with cellular and anatomical features [4]. These findings support looking for shared and region-specific molecular patterns; reproducibility of anatomical signatures, however, does not directly quantify whether one donor's regional expression profile can predict another region in a new donor.

GTEx complements spatial atlases by sampling multiple tissues across many individuals. The consortium used bulk RNA-seq and genetic association analyses to map shared and tissue-specific regulatory effects [5]. Its matched-donor design makes cross-region modelling feasible, but regional missingness and heterogeneous tissue composition complicate interpretation. Kim-Hellmuth *et al.* combined inferred cell-type enrichment with genotype to characterize cell-type interaction effects on expression [6]. Their results show why apparently shared or tissue-specific regulatory signals can change when cellular composition is considered. The present study therefore treats cell mixture as a plausible alternative explanation for predictive patterns, rather than interpreting every performance difference as a distinct neuronal program.

The closest computational precedent is BRONTE, which integrates GTEx and the Allen Brain Atlas using learned gene representations and brain-ontology information to impute expression in additional tissues [7]. It evaluates held-out expression measurements and the recovery of biological patterns. This demonstrates the promise of incorporating anatomical information into expression prediction, but its multi-region inputs and ontology-based model differ from a directional, single-source-region comparison. The version cited here is a preprint, and its results will be interpreted with that status explicit. This project does not reproduce BRONTE or perform its downstream genetic-risk analyses.

A manageable gap is therefore the need for an interpretable comparison of directional cross-region prediction under a consistent evaluation procedure, coupled with a cautious account of which molecular characteristics accompany differences in predictability. The project will investigate this gap using a regularized baseline model, training-only feature selection and evaluation on unseen donors. It does not claim that cross-tissue prediction is new. Its contribution is the particular comparison and analysis of molecular features, including transparent reporting of negative results and sensitivity to donor cohorts and target-gene selection.

The disease-gene component uses a deliberately small literature-defined panel: six established monogenic Parkinson's genes, three familial Alzheimer's genes, and four commonly implicated ALS genes [8]-[10]. These panels provide biological context, but are neither exhaustive disease signatures nor direct measurements of regional susceptibility. Differences in their prediction scores will generate hypotheses for follow-up, not demonstrate disease mechanisms.

3. Dataset and Responsible Use

The dataset consists of Adult GTEx V11 regional gene-read-count matrices supplied for this project and inspected on October 9, 2026. The public source is the GTEx Portal [11]. The V11 portal documentation describes an annotation update to GENCODE 47. The consortium atlas publication [5] establishes the broader resource but reports V8, not these V11 matrices. The original download date and the exact V11 export/release provenance will be recorded by the team; filenames alone do not establish every processing detail.

Thirteen compressed GCT 1.2 files were inspected. Each contains 74,628 annotated gene rows, with Ensembl identifiers, gene descriptions and sample-specific read counts. Across all files there are 3,234 tissue-sample columns and 393 distinct donor identifiers. The main analysis excludes cervical spinal cord, leaving 12 brain-region files, 3,030 tissue records and 385 unique donors. Within each supplied region, every sample maps to a distinct donor; the same donor can appear in several regions. These counts were computed from file headers and sample identifiers, not inferred from published GTEx cohort totals.

Table 1. Sample coverage in the supplied matrices. Sample and unique-donor counts are identical within each region. Totals across regions count repeated donors more than once.

Region	Samples / donors	Main analysis
Amygdala	181	Included
Anterior cingulate (BA24)	233	Included
Caudate	300	Included
Cerebellar hemisphere	277	Included
Cerebellum	266	Included
Cortex	270	Included
Frontal cortex (BA9)	269	Included
Hippocampus	255	Included
Hypothalamus	257	Included
Nucleus accumbens	285	Included
Putamen	254	Included
Spinal cord (cervical C-1)	204	Excluded
Substantia nigra	183	Included

One independent case is a donor. For a region pair, the analytical record comprises the source and target expression profiles from that donor. Outcomes are normalized target-gene expression and held-out prediction performance. Eligibility requires measurements in both regions and at least 60 paired donors. Genes must pass training-set expression and variance checks. Missing regions are not imputed for the initial analysis.

GTEx distinguishes open-access processed expression information from protected information, including raw data and attributes that may identify donors [12]. The project uses processed counts, does not attempt re-identification, and does not require protected genotype data. Public project materials will contain code and aggregate outputs; donor-level split/error files and original matrices will be kept out of the repository while the team verifies applicable sharing conditions. Source links, file hashes and access records will support reproducibility without unnecessary redistribution.

The tissue is postmortem bulk material, not a PD/AD/ALS case-control cohort or a collection of purified neurons. Age, sex, ancestry, tissue quality, postmortem interval and cell mixture may affect prediction. These covariates are absent from the matrices. Permitted metadata will be sought; limited adjustment and population generalizability will be acknowledged.

4. Proposed Methods and Evaluation

The current Google Colab notebook provides a working initial pipeline. It reads GCT dimensions and numeric values, checks nonnegative finite counts and nonzero library totals, and derives donor identifiers from sample names. Counts are converted to counts per million using the full per-sample matrix, then transformed as log2(CPM + 1). This is an exploratory normalization and is not TPM. Because library-size scaling may retain composition effects, a normalization sensitivity analysis, for example with TMM, will be added before final interpretation [13].

For each directional pair, matched donors are sorted and split reproducibly into 80% development and 20% test sets (seed 42). The development set is split again into 75% inner training and 25% validation. No test donor is used for selection, scaling, fitting or penalty tuning. All measurements contributing to a pairwise case remain together. Within-region replicates, if encountered in future exports, will be aggregated after sample-level normalization so the model contains one record per donor and region.

Using inner-training donors only, genes must exceed CPM 1 in at least 20% of donors and have nonzero variance. The model selects 500 source predictors and 1,000 target genes ranked by training variance. Eligible prespecified disease-panel genes are added to the targets, yielding 1,012-1,013 targets in the pilot. Input scaling is fitted on training donors. Multi-output ridge regression is evaluated at penalties 1, 10, 100 and 1,000, with validation mean squared error selecting the penalty [14]. After selection, scaling and regression are refitted on all development donors while the target/predictor lists remain fixed. This separation prevents using the evaluation set to optimize the model [15].

The primary result will be the median gene-wise test R^2 for each source-to-target direction. For gene *g*, $R^2 = 1 - SSE_g/SST_g$, where SST is variation around that gene's mean in the test cohort. Negative values are retained and constant genes are reported as undefined. This measure assesses donor-level prediction for individual genes, avoiding a pooled correlation dominated by differences in average abundance across genes. It quantifies performance for the selected targets under this design, not a causal degree of molecular sharing.

A development-set target-mean predictor supplies the practical baseline. A complementary prediction skill score is $1 - MSE_{model}/MSE_{baseline}$, averaged across target genes and test donors. Skill zero means matching that baseline, positive values improve it, and negative values perform worse. This score differs from conventional test R^2 because it uses a development-set reference mean. Conditional 95% intervals will be estimated with 1,000 bootstrap resamples of test donors using fixed predictions, including a median-gene-R^2 interval. These intervals do not incorporate model refitting or training uncertainty.

Molecular analyses will relate gene-wise predictability to development-set target expression, target variance and the absolute source-target difference in mean expression using descriptive Spearman associations. Disease-panel gene R^2 will be displayed with eligibility/coverage tables [8]-[10]. Genes are correlated, and panel membership alone is not mechanistic evidence; the pilot will not assign independent-gene p-values or label this analysis pathway enrichment.

The pilot examines six directions; the final analysis can extend to all 132 ordered pairs among the 12 regions. Before interpreting rankings, the team will repeat donor splits, compare common target genes and, where sufficient, common donor cohorts, and evaluate metadata/normalization sensitivity. These are planned extensions, not completed validations. Python, pandas, NumPy, scikit-learn and Matplotlib perform loading, modelling and visualization. The notebook exports dataset summaries, pair-level scores, gene-level characteristics, figures and configuration records; the final repository will document these outputs and the software versions.

5. Preliminary Results

A local execution of the AI-assisted notebook on the supplied files completed six directional models. The paired cohorts contained 194-233 donors, with 39-47 donors held out per direction (Table 2). The matrices passed the notebook's dimension, nonnegative/finite-value and nonzero-library checks. Table 1 and Figure 1 establish adequate matched regional coverage for a pilot while showing that sample availability is unequal.

Table 2. Verified pilot performance. R^2 is the median of gene-wise held-out-donor scores. Skill uses the development-mean baseline. Each direction evaluates approximately 1,000 training-selected targets plus eligible disease-panel genes.

Prediction direction	Paired / test	Median R^2	Skill
Caudate to Putamen	233 / 47	0.450	0.457
Putamen to Caudate	233 / 47	0.385	0.347
Cortex to Frontal cortex (BA9)	194 / 39	0.496	0.472
Frontal cortex (BA9) to Cortex	194 / 39	0.363	0.386
Cerebellum to Cerebellar hemisphere	222 / 45	0.294	0.379
Cerebellar hemisphere to Cerebellum	222 / 45	0.272	0.346

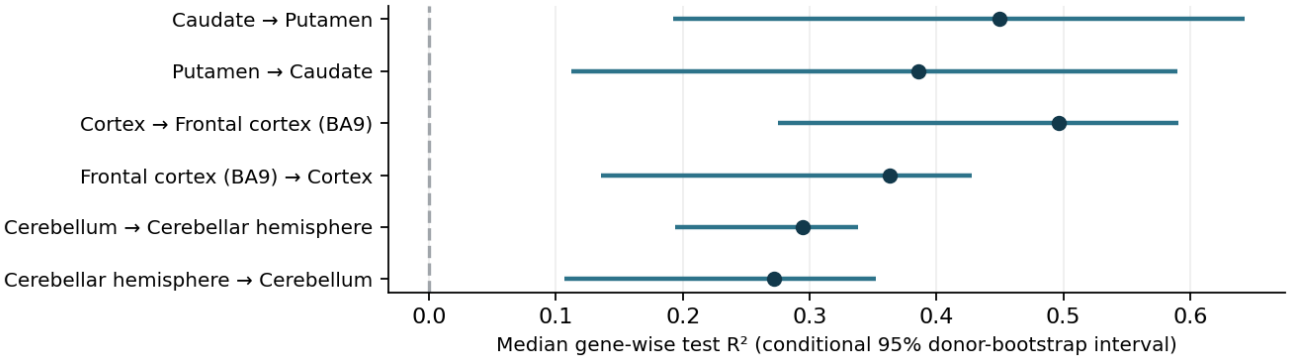


Figure 2. Median gene-wise test R^2 and conditional 95% intervals from 1,000 donor-bootstrap resamples of fixed predictions. Overlapping intervals, different cohorts and different targets preclude a definitive regional ranking.

Median gene-wise R^2 ranged from 0.272 to 0.496; skill ranged from 0.346 to 0.472. The pilot therefore demonstrates that the pipeline can recover target-region variation beyond the simple mean baseline for these selected genes and splits. The direction-dependent scores suggest that transferability is not symmetric under this model, but they do not establish intrinsic biological asymmetry because target composition and variance also differ.

For caudate-to-putamen, regional mean-expression difference had a descriptive Spearman association of -0.376 with gene-wise R^2; the reverse direction was -0.408. The association was weaker or absent in other pilot directions, including cortex-to-frontal cortex (0.013). All 13 prespecified disease genes were evaluated in each pilot direction. These observations motivate broader and controlled comparisons, rather than a universal conclusion about regional differences or vulnerability.

The pilot supports retaining the RQ with a precise definition of predictive transferability and an explicitly exploratory disease component. The largest remaining uncertainty is whether differences persist when target genes, cohorts, technical factors and cell composition are controlled. Final work must add repeated splits and sensitivity analyses, document metadata and data provenance, and verify literature-to-gene interpretations. The dataset can support predictive and descriptive molecular associations; it cannot establish disease causation, selective neuronal death or clinical validity.

6. Teamwork and Project Plan

The proposed allocation below separates biological interpretation, data stewardship and modelling while requiring joint review. Role labels must be replaced with team-member names before submission. For a three-person group, the analysis lead also coordinates reproducibility; for four members, that role can be assigned separately. These assignments are proposed responsibilities, not claims of completed contributions.

Table 3. Proposed milestones and owners through the final submission.

Dates	Milestone / evidence	Proposed owner
Oct. 9-16	Review citations and pilot outputs; complete midterm; publish working notebook/README and confirm access.	Literature lead; data lead; analysis lead
Oct. 17-30	Record V11 provenance; obtain permitted covariates; check missingness and normalization sensitivity.	Data lead; analysis lead
Oct. 31-Nov. 13	Run eligible pairs and repeated splits; test common-gene/cohort comparisons; report baseline and uncertainty.	Analysis lead; reproducibility lead
Nov. 14-20	Check molecular associations and gene-panel coverage; compare findings with literature and alternative explanations.	Literature lead; analysis lead
Nov. 21-27	Draft paper, captions and limitations; independently rerun central output and review repository instructions.	All members
Nov. 28-Dec. 4	Revise paper; rehearse presentation; verify links; freeze graded repository commit and submit.	All members

Data and Code Availability

Dataset source: https://gtexportal.org/home/datasets/. Working repository: **[INSERT WORKING GITHUB REPOSITORY URL]**. The existing notebook is *GTEx_V11_Cross_Region_Analysis.ipynb*. The repository must include a README with the RQ, source/file access instructions, Colab setup, folder map and steps to reproduce Table 1 and Figure 2. Proposed folders are *notebooks/*, *results/* and *figures/*; original matrices and donor-level PRIVATE outputs are excluded pending sharing review. The notebook currently generates aggregate CSVs and PNGs and can export them together.

AI-Assisted Work

ChatGPT assisted with literature retrieval, notebook development, debugging, preliminary data analysis, graph generation and drafting this proposal. The analysis code was executed on the supplied files, and the plotted pilot results were checked against the resulting tables during this session. This does not substitute for team verification. Before submission, team members must inspect cited sources, review donor-level splitting and training-only preprocessing, rerun at least a central output, and check that the written claims match it. **Team verification completed:** [INSERT ACTUAL CHECKS, MEMBERS AND DATE].

Author Contributions

[Member A]: [Record actual research-question, literature and writing contributions]. **[Member B]:** [Record actual data-preparation, responsible-use and analysis contributions]. **[Member C]:** [Record actual modelling, code, figures and evaluation contributions]. **[Member D, if applicable]:** [Record actual reproducibility, interpretation and revision contributions]. All authors should confirm the final text and their recorded roles. Do not submit proposed responsibilities as completed work.

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